MIL-C-915/32D
15 December 1976
SUPERSEDING
MIL-C-915/32C
23 August 1976

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 1000 VOLTS,

TYPE MSCU

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the cable described herein shall consist of this document and the latest issue of Specification MIL-C-915.

REQUIREMENTS:
Qualification required.

Construction (Watertight)

First - Copper conductor, uncoated, Navy Standard size 2(7).
Second - Extruded silicone rubber insulation. Nominal (nom) thickness 0.018 inch. Diameter over insulation 0.084-inch minimum (min).
Third - Glass braid.
Fourth - Braid covering. Standard Identification Code, applied by method 1. Diameter over braid covering 0.113-inch maximum (max).
Fifth - The required number of conductors (see Details) cabled together with a lay not greater than 24 times the pitch diameter of the layer, except the lay of the two outer layers for sizes 44, 61, and 91 shall not exceed 18 times the pitch diameter of the layer. Non-fibrous fillers shall be used to form a firm, well-rounded assembly. Cabling sequence shall be consecutive, starting with No. 1, from the center outward.
Sixth - Binder or combination binder/barrier.
Seventh - Special thermoplastic jacket (gray) (see Details for thickness). Cable surface marking required.

Note 1: When this cable is manufactured with armor, the following additional requirements shall apply:

(D)
(a) The type designation shall change to MSCA.
(b) Standard thermoplastic jacket (black or gray) may be used.
(c) Braided metal armor and paint shall be added.
(d) Cable surface marking will not be required.
(e) The maximum overall dimension shall be increased by 0.050-inch.
(f) Group C inspection shall include: Armor - Conformance to material, construction and coverage. When standard thermoplastic jacket is used, the air oven test does not apply.

(D) denotes changes

FSC 6145

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T12:09Z
Check the source to verify that this is the current version before use.

Details

Type and size	Number of condrs.	Jacket thickness nom (inch)	Overall diameter max (inches)
MSCU-7	7	0.040	0.484
MSCU-10	10	.050	.622
MSCU-14	14	.050	.668
MSCU-19	19	.050	.738
MSCU-24	24	.050	.855
MSCU-30	30	.050	.901
MSCU-37	37	.060	1.002
MSCU-44	44	.060	1.114
MSCU-61	61	.060	1.250
MSCU-91	91	.060	1.480

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Voltage withstand - Volts, root mean square (Vrms), min	3000
Insulation resistance - Megohms/1000 ft, min	500
Conductor resistance - Ohms/1000 ft, At 25°Celsius (C), max	6.44
Group A:	
Visual and dimensional	
Watertightness - See MIL-C-915, table of limits for water leakage	250
Abrasion resistance - Scrapes, minimum	No damage
Crack resistance	
Group B:	
Gas flame - 1 hour	No failure
Drip - At 95°C ± 1°C	Zero
Cold bending, cable - (Max mandrel diameter 12 times cable diameter)	No damage
Physicals (unaged)	
Insulation	
Tensile strength - pound-force per square inch (lbf/in^2), min	750
Elongation - Percent, min	125
Jacket (Standard thermoplastic)	
Tensile strength - lbf/in^2, min	1800
Elongation - Percent, min	225
Jacket (Special thermoplastic)	
Tensile strength - lbf/in^2, min	2300
Elongation - Percent, min	250
Group C:	
Physicals (aged)	
Jacket (Standard thermoplastic)	
Hot oil immersion	
Tensile strength - lbf/in^2, min	1400
Elongation - Percent, min	50
Jacket (Special thermoplastic)	
Air oven (168 hours at 136°C)	
Tensile strength - Percent of unaged, min	80
Elongation - Percent of unaged, min	60
Hot oil immersion	
Tensile strength - Percent of unaged, min	75
Elongation - Percent of unaged, min	50
Permanence of printing (conductor) - Cycles, min	50
Permanence of printing (jacket) - Cycles, min	250
Cable filler removability	

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T12:09Z
Check the source to verify that this is the current version before use.

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of group A, B and C plus the following:

Cable aging and compatibility (125°C ± 5°C)	No failure
Cold working (minus 20°C ± 2°C)	No damage
Gas flame (3 hours)	No failure
Specific gravity of extruded silicone insulation, max	1.55

UNIT ORDERING LENGTHS:

Type and size	Feet (nom)
MSCU-7 and MSCU-10	1500
All other sizes	1000

Custodians:
Army - MI
Navy - SH
Air Force - 99

Review activities:
Army - EL, AV
Navy - EC

User activities:
Army - ME, MU, WC
Navy - CG

Preparing activity:
Navy - SH

(Project 6145-0708)

★U.S. GOVERNMENT PRINTING OFFICE: 1977— 703-020:790

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T12:09Z
Check the source to verify that this is the current version before use.